GeoFusionNet / GeoCorr: Status Report

Status as of 2026-10-11, 18:30 UTC (00:00 IST 10-12) · main @ e0c5337 (+ uncommitted --settings filter in scripts/fusion_sweep.py) · Prepared for Gobika S

> **Conventions.** DTU test, 22 scans, 1600×1152, official protocol. Values are in mm; lower is better. Every "±" in this report is the population standard deviation over seeds (**ddof = 0**), for A0f and C1 alike. **PENDING** marks a number that has not been measured yet.

1. Status

Item	State	Expected
C1 3-seed eval (B2_capacity s0, s1, s2)	**Done** 2026-10-11 07:15 UTC	n/a
Track 1 screen (B2_mono_feat s2)	**Done** 2026-10-11 11:07 UTC	n/a
Fusion sweep, full grid (A0f s2 vs C1 s2) scripts/queue_fusion_sweep.sh · runs/fusion_sweep/	**Stopped** 2026-10-11 18:17 UTC after 6 of 44 model-scans (scan1, scan4, scan9 × 2 models); partial outputs kept. It ran at ~3.3–4 h per model-scan (projected end ~10-12 23:30 IST) and slowed training by ~2.4%.	n/a
Fusion sweep, reduced (distance only, n = 4) scripts/queue_fusion_sweep_dist_n4.sh · runs/fusion_sweep_dist_n4/	Running since 18:18 UTC, CPU only, 6 workers × 8 threads; the 6 finished model-scans are reused	**~2026-10-11 22:00 UTC (03:30 IST 10-12), estimate**
B2_nogeo seed 0 (L_geo off, full chain) scripts/queue_b2_nogeo_seed0.sh · log runs/queue_b2_nogeo_seed0.log	Base training, step 10000/20000 at 18:17 UTC (2.10 s/step vs 2.05 on seed 2); then finetune 6k → final 1.5k → diagnostics → 1600×1152 eval (23 h 27 m on seed 2)	**~2026-10-12 17:15–18:30 IST (PENDING)**
C1 placebo prior (E)	Prepared, **not launched** (by decision)	n/a

If B2_nogeo seed 0 has not been evaluated by 20:00 IST on 2026-10-12, this report is finalised with L_geo at n = 1.

2. Headline: C1 vs Stage 0 (3 seeds each)

Model	Acc	Comp	Overall
A0f seed 0	0.436	0.558	0.497
A0f seed 1	0.441	0.555	0.498
A0f seed 2	0.435	0.524	0.479
A0f (Stage 0), mean ± std (ddof=0)	0.437 ± 0.003	0.545 ± 0.016	**0.491 ± 0.009**
C1 seed 0	0.470	0.358	0.414
C1 seed 1	0.487	0.373	0.430
C1 seed 2	0.474	0.362	0.418
C1 (B2_capacity), mean ± std (ddof=0)	0.477 ± 0.008	0.364 ± 0.006	**0.421 ± 0.007**
C1 − A0f	+0.040	−0.181	−0.071

C1 is better overall by 0.071 mm, about 14%, and the seed ranges do not overlap (C1 0.414–0.430, A0f 0.479–0.498). This is a trade, not a uniform gain. Completeness improves by 0.18 mm and accuracy gets 0.04 mm worse. The worst scans are the same as A0f's: scan48 (0.71–0.77), scan77, scan11, scan13 and scan49. On seed 2, C1 beats A0f on 17 of 22 scans. CL-MVSNet is not compared here. Our 0.438 number for CL-MVSNet comes from point clouds generated locally at 640×480, and the cause of its gap to the published 0.329 is unconfirmed.

3. Does C1's completeness gain survive a looser or stricter fusion? (reduced fusion sweep)

Both models' depth maps (seed 2 each, already dumped at 1600×1152) are re-fused and scored with the official metric. The full 32-setting grid was too slow (~3.3–4 h per model-scan), so it was stopped and replaced by a reduced sweep.

Rule, fixed before any result was looked at: all 22 scans; family *geo* (world-distance check) at distance 0.1 / **0.2** / 0.4 / 0.8 mm; consistent-view count fixed at the official 4; prob 0.3. That is the official setting (0.2 mm), one stricter and two looser. Estimated ~30–35 min per model-scan, 38 to run.

> **What this sweep can support:** whether C1's completeness gain and accuracy loss over A0f depend on the fusion *distance* threshold, and a matched-point-count comparison along that one axis, for scans whose point counts overlap.
> **What it cannot support:** any conclusion about the view-count threshold (2/3/6 views not run); any conclusion about MVSNet-style reprojection/relative-depth filters (not run); anything about the gap to CL-MVSNet (different checkpoint and resolution, not part of this sweep). It is single-seed (seed 2 of each model).

Check	A0f s2	C1 s2	Note
Reproduction of the official setting (geo 0.2 mm, 4 views) against eval_1600x1152.json, max |ΔOverall| over all 22 scans	**PENDING**	**PENDING**	Must be ~0
Fused points, official setting (mean, M)	**PENDING**	**PENDING**	
Comp at A0f's point count	**PENDING**	**PENDING**	C1 better on **PENDING** scans in range
Comp at C1's point count	**PENDING**	**PENDING**	C1 better on **PENDING** scans in range
Acc / Overall at matched counts	**PENDING**	**PENDING**	

Verdict: PENDING. The per-setting table will be in runs/fusion_sweep_dist_n4/report.log. The full-grid results for scan1, scan4 and scan9 stay in runs/fusion_sweep/ and are not used for the verdict, so all 22 scans are treated the same.

4. Track 1 screen: frozen mono features (B2_mono_feat seed 2)

Run (seed 2)	Acc	Comp	Overall
B2_nogeo (matched control)	0.461	0.410	0.436
B2_mono_feat	0.464	0.382	0.423
mono_feat − control	+0.003	−0.029	**−0.013**

Screening result, not a confirmed gain.

Matched setup (checked against configs and logs):

- Both runs are initialised from **runs/B2_nogeo_finetune/seed2/ckpt_final.pt** (the "initialized weights from" line is in both train.log files). Everything upstream of the final stage is the same checkpoint.
- Same final-stage budget: 1500 steps each, lr 5e-4, same data, batch and loss settings. The config diff (B2_nogeo.yaml vs B2_mono_feat.yaml) shows only the mono-feature input (frozen Depth-Anything-V2-Small, zero-initialised injection at coarse and mid), data.geometry_prior (mono_sparse vs none) and use_geo_term (true vs false).

- **L_geo is off in both:** geo_weight 0.0 in both; the control has use_geo_term true with weight 0. B2 never reads the prior input, so the geometry_prior and use_geo_term differences do not change the computation.
- One budget difference: mono_feat has 12,480 more trainable parameters (2,133,939 vs 2,121,459, the injection layers), plus the frozen Depth-Anything model. Step time is about the same (4.17 vs 4.21 s/step).

Result:

- ΔOverall −0.0127 (mono_feat − control), larger than 0.009.
- Better on only 11 of 22 scans.
- scan48 alone accounts for −0.155 (0.870 → 0.715). Without scan48, the mean difference over the other 21 scans is −0.006, below 0.009.
- Single seed.
- The gain is from completeness (−0.029); accuracy is slightly worse (+0.003).

We read this as a screening signal concentrated on the ambiguous scan, not a confirmed gain. A second seed would be needed before Track 1 counts as a result.

Per-scan Overall (mm), seed 2

Scan	Control (B2_nogeo)	B2_mono_feat	ΔAcc	ΔComp	ΔOverall
scan1	0.280	0.281	+0.002	+0.000	+0.001
scan4	0.335	0.343	+0.008	+0.008	+0.008
scan9	0.348	0.357	+0.014	+0.004	+0.009
scan10	0.298	0.303	+0.009	+0.001	+0.005
scan11	0.681	0.617	+0.003	−0.131	−0.064
scan12	0.339	0.342	+0.011	−0.006	+0.003
scan13	0.521	0.510	+0.026	−0.048	−0.011
scan15	0.323	0.334	+0.017	+0.006	+0.011
scan23	0.311	0.321	+0.014	+0.006	+0.010
scan24	0.325	0.344	+0.015	+0.022	+0.019
scan29	0.433	0.419	−0.025	−0.003	−0.014
scan32	0.499	0.442	−0.004	−0.109	−0.057
scan33	0.491	0.491	+0.004	−0.005	−0.000
scan34	0.243	0.239	−0.004	−0.004	−0.004
scan48	0.870	0.715	−0.006	−0.305	**−0.155**
scan49	0.589	0.606	+0.001	+0.033	+0.017
scan62	0.570	0.543	−0.027	−0.027	−0.027
scan75	0.381	0.377	+0.012	−0.020	−0.004
scan77	0.740	0.732	+0.012	−0.028	−0.008
scan110	0.397	0.377	−0.016	−0.024	−0.020
scan114	0.261	0.264	+0.003	+0.003	+0.003
scan118	0.347	0.348	+0.001	−0.000	+0.001
Mean (22)	0.436	0.423	+0.003	−0.029	**−0.013**

Δ = mono_feat − control; negative is better. mono_feat is better on 11 of 22 scans.

5. L_geo on/off: C1 vs B2_nogeo (full chain, L_geo off in every phase)

> **L_geo: n = 1, second seed pending.**

Seed	Run	Acc	Comp	Overall	Per-scan Overall wins (of 22)
2	C1 (L_geo on, geo_weight 0.5)	0.474	0.362	0.418	11
	B2_nogeo (L_geo off)	0.461	0.410	0.436	11
	off − on	−0.013	+0.048	+0.018	
0	C1 (L_geo on)	0.470	0.358	0.414	**PENDING**
	B2_nogeo (L_geo off)	**PENDING**	**PENDING**	**PENDING**	**PENDING**
	off − on	**PENDING**	**PENDING**	**PENDING**	
Paired diff, mean ± std (ddof=0), n = 2		**PENDING**	**PENDING**	**PENDING**	same sign in both pairs: **PENDING**

On seed 2, turning L_geo off makes overall 0.018 mm worse. That would mean L_geo helps, but the effect is mixed:

- All of it comes from completeness (+0.048).
- Accuracy is 0.013 better with L_geo off.
- The scan wins split 11 / 11.

With one seed we cannot tell this apart from seed-to-seed variation. C1's own seed spread is 0.414–0.430. The seed 0 pair decides whether "L_geo helps C1" can be claimed.

6. Not run, by decision

- C1 placebo prior (blurred, noisy and cross-scene G), configs prepared; patches/C1_placebo.patch not applied.
- No other GPU jobs until B2_nogeo seed 0 finishes.